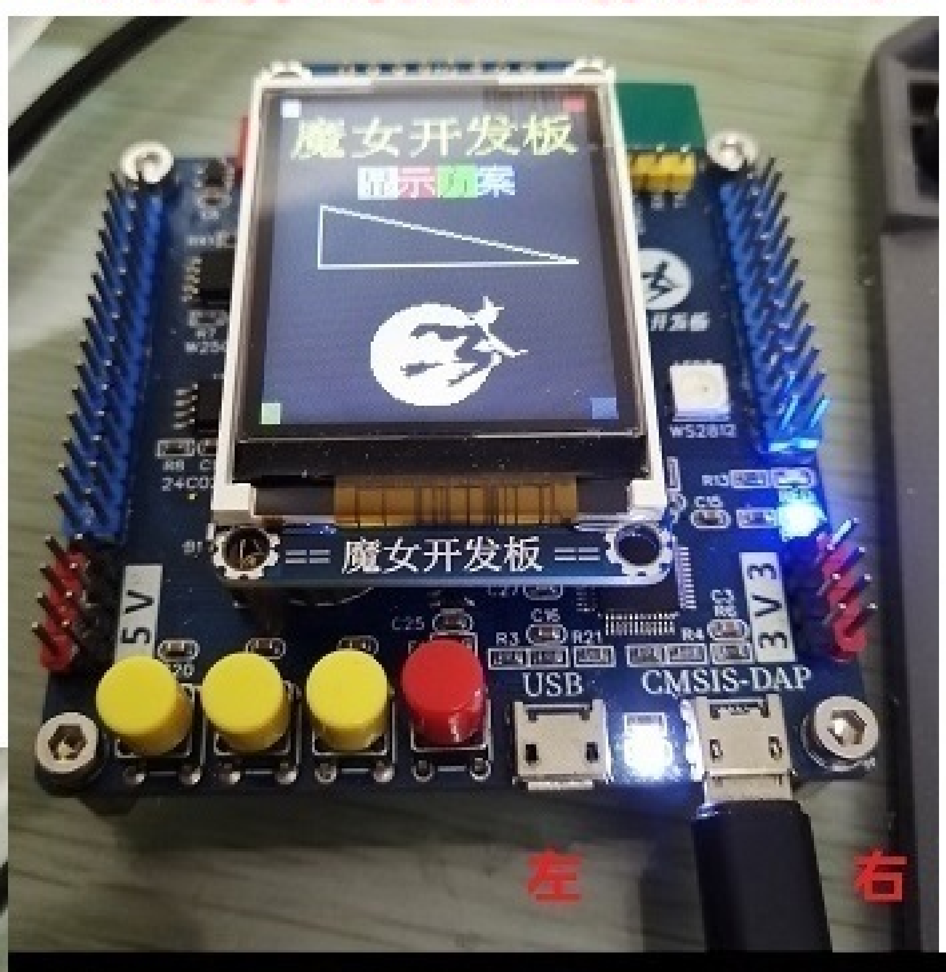
****** 本篇为长图注解，注意可向下移动查看 ******

# 1:数据线的连接方法:



(两个USB接口的区别)

**左侧接口：** 用户自定义编程接口

**右侧接口：** 1-供电 5V-400mA
2-烧录下载 CMSIS-DAP
3-仿真调试
4- 虚拟串口 UART1

(连接后状态指示)

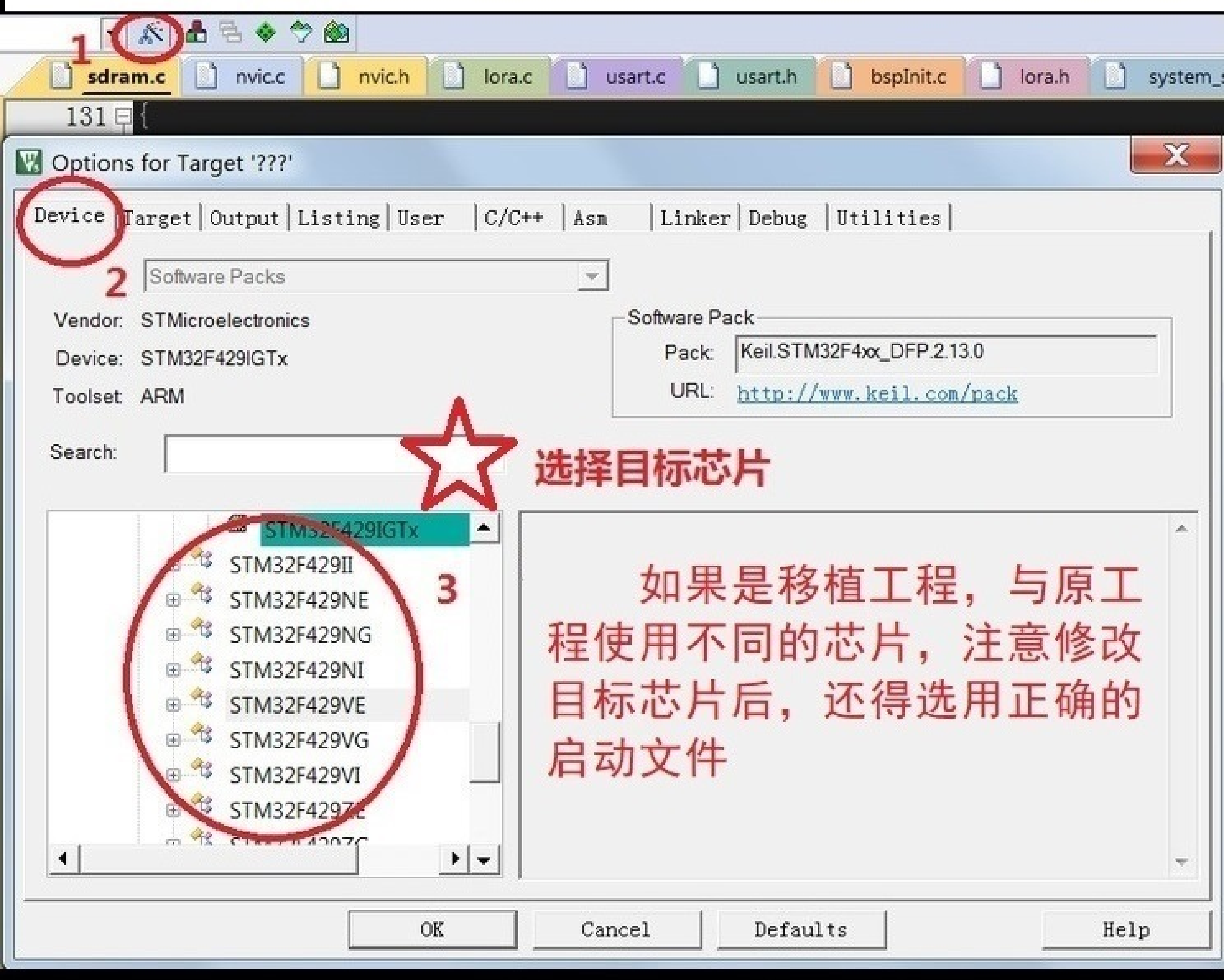
**白色LED：** 规律闪烁 - 连接成功
只亮不闪 - 连接失败

使用安卓数据线，连接右下角的CMSIS-DAP接口至电脑

1、CMSIS-DAP调试器：不用安装驱动，不管是WIN7还是WIN10；连接电脑后，Micro USB接口旁边的LED就会闪烁，表示正常工作中，无闪烁表示没有连接电脑；

2、虚拟串口：win10免驱，win7需要安装驱动(不安装不影响CMSIS-DAP调试器的使用)；
硬件已连接PA9\PA10的USART1，按正常USART通信收发即可；

# 2: 下载器参数设置步骤(只需设置1次)



Options for Target '???'

Software Packs

Vendor: STMicroelectronics
Device: STM32F429IGTx
Toolset: ARM

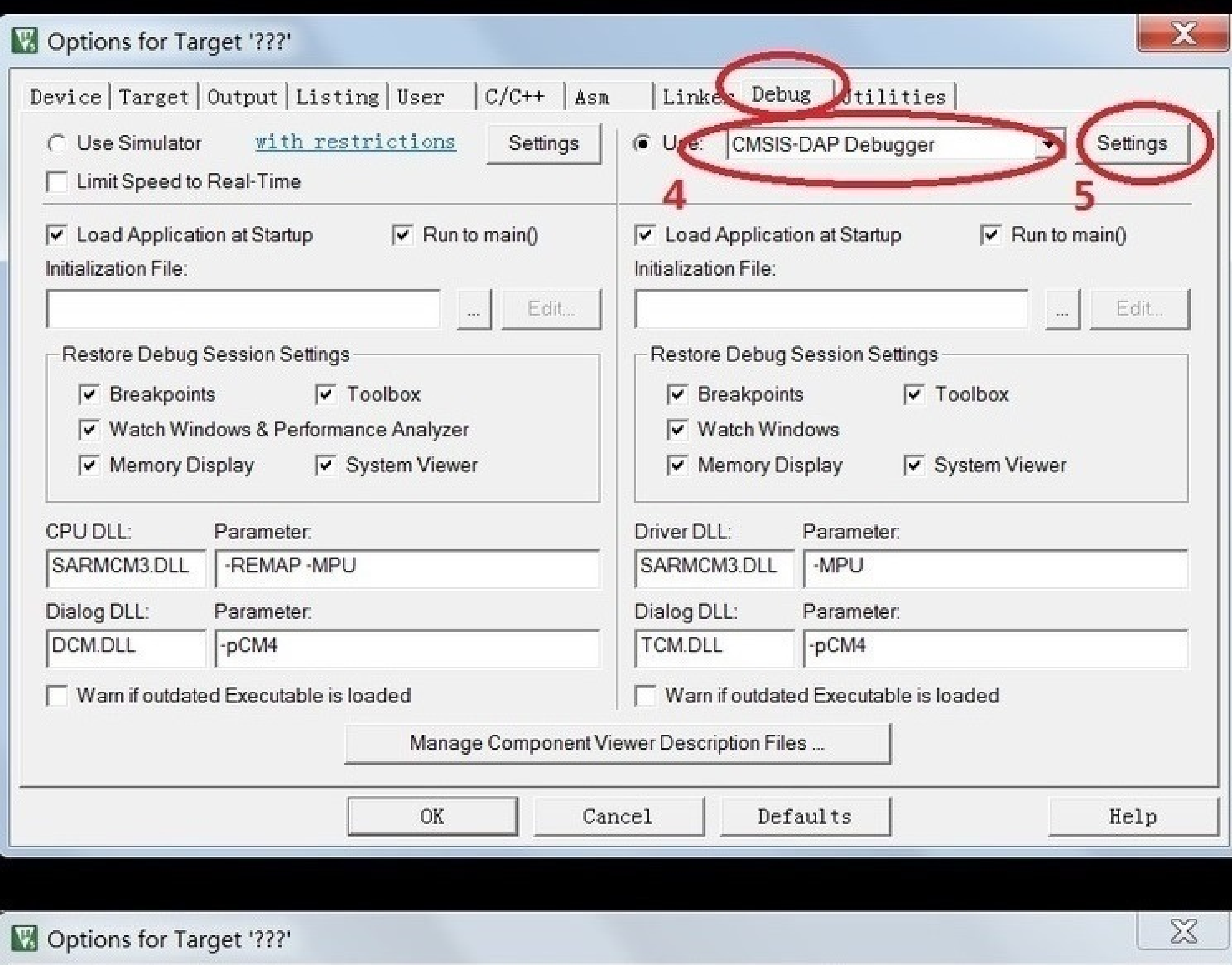
Software Pack
Pack: Keil.STM32F4xx_DFP.2.13.0
URL: http://www.keil.com/pack

Search:

STM32F429IGTx
STM32F429II
STM32F429NE
STM32F429NG
STM32F429NI
STM32F429VE
STM32F429VG
STM32F429VI
STM32F429ZE

选择目标芯片

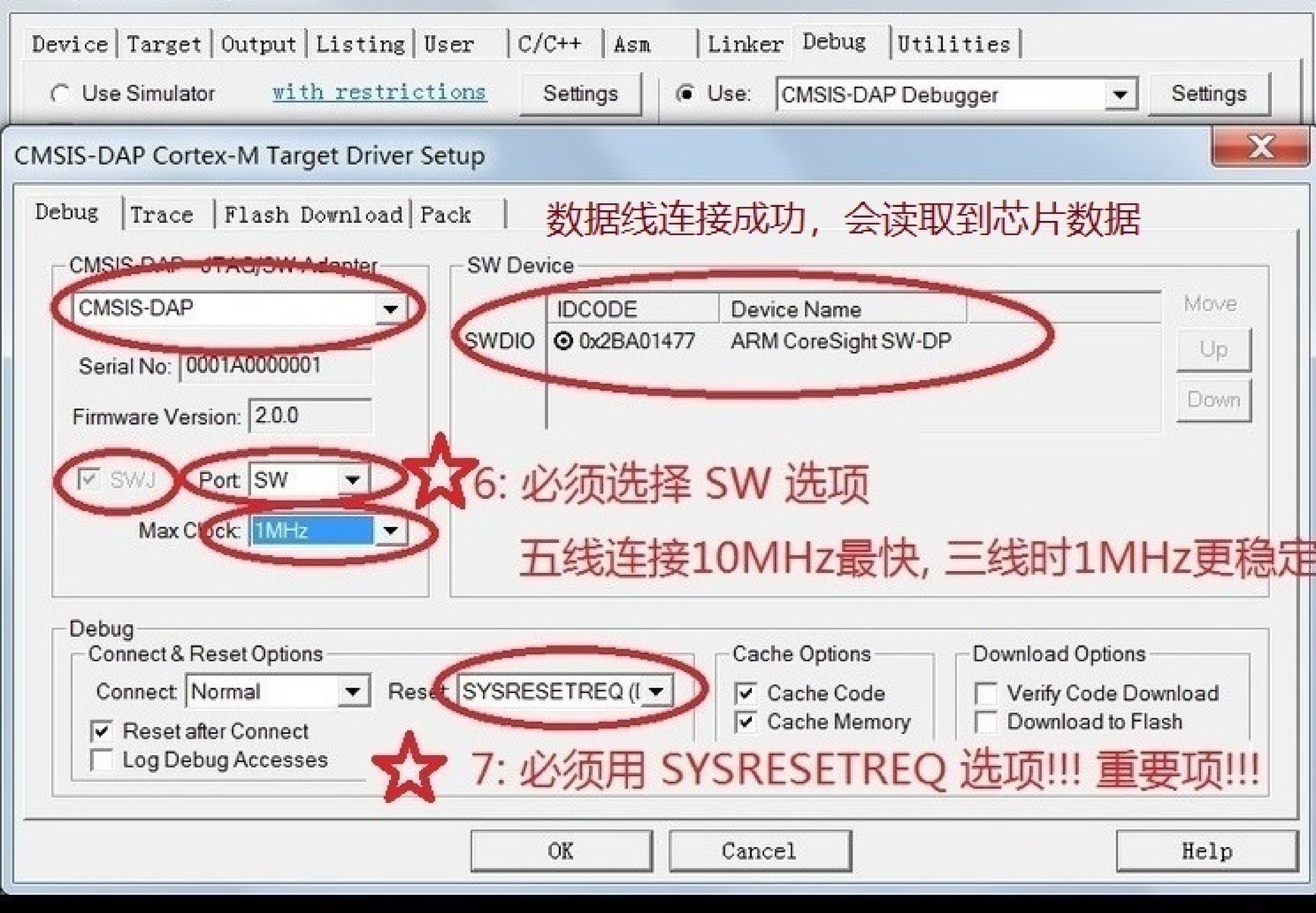
如果是移植工程，与原工程使用不同的芯片，注意修改目标芯片后，还得选用正确的启动文件



Options for Target '???'

○ Use Simulator with restrictions
☐ Limit Speed to Real-Time
☑ Load Application at Startup ☑ Run to main()
Initialization File:
Restore Debug Session Settings
☑ Breakpoints ☑ Toolbox
☑ Watch Windows & Performance Analyzer
☑ Memory Display ☑ System Viewer
CPU DLL: SARMCM3.DLL Parameter: -REMAP -MPU
Dialog DLL: DCM.DLL Parameter: -pCM4
☐ Warn if outdated Executable is loaded

● Use: CMSIS-DAP Debugger
☑ Load Application at Startup ☑ Run to main()
Initialization File:
Restore Debug Session Settings
☑ Breakpoints ☑ Toolbox
☑ Watch Windows
☑ Memory Display ☑ System Viewer
Driver DLL: SARMCM3.DLL Parameter: -MPU
Dialog DLL: TCM.DLL Parameter: -pCM4
☐ Warn if outdated Executable is loaded



Options for Target '???'

○ Use Simulator with restrictions ● Use: CMSIS-DAP Debugger

CMSIS-DAP Cortex-M Target Driver Setup

数据线连接成功，会读取到芯片数据

CMSIS-DAP - JTAG/SW Adapter: CMSIS-DAP
Serial No: 0001A0000001
Firmware Version: 2.0.0
☑ SWJ Port: SW
Max Clock: 1MHz

SW Device

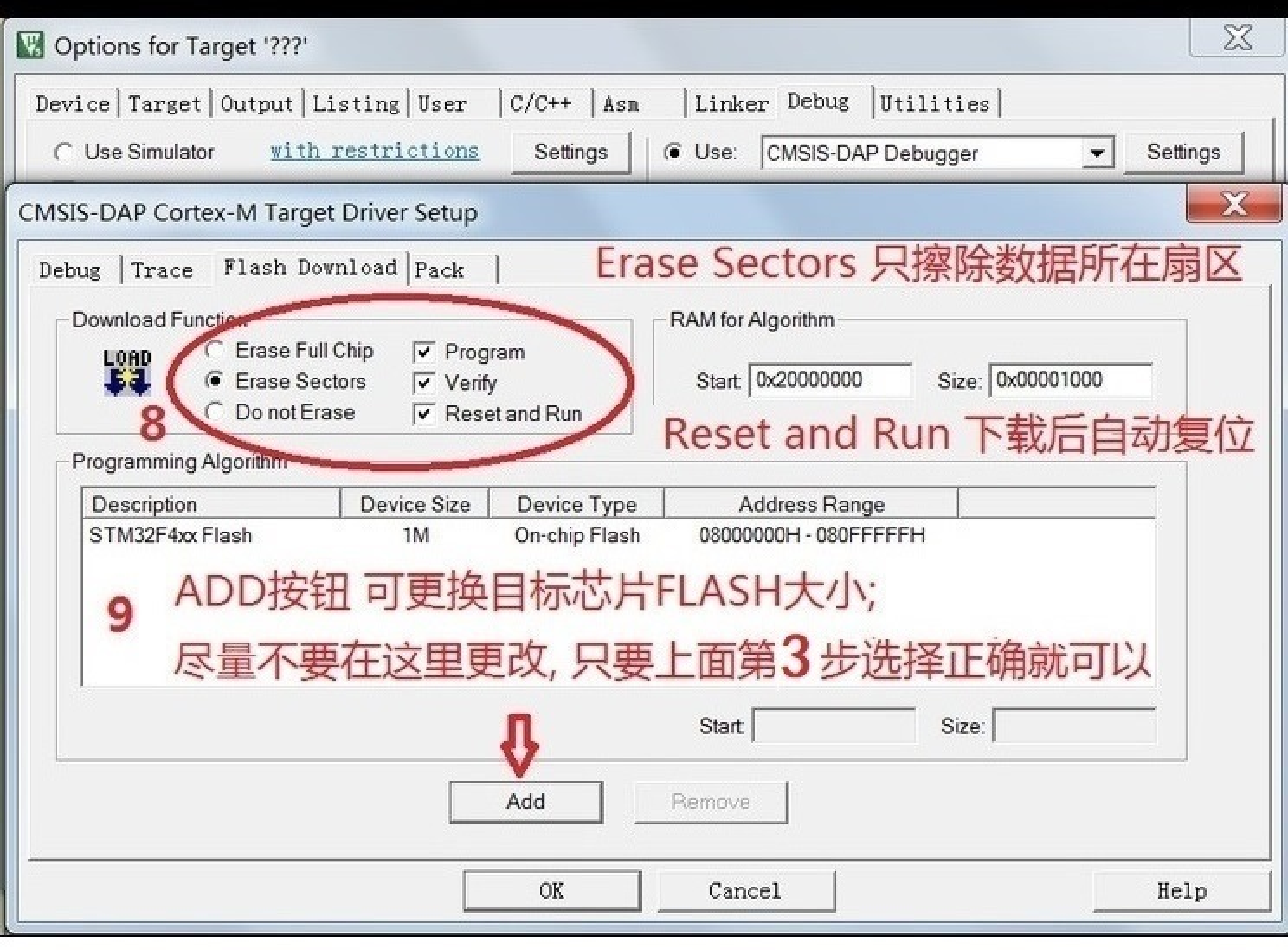
| | IDCODE | Device Name |
|---|---|---|
| SWDIO | ⊙ 0x2BA01477 | ARM CoreSight SW-DP |

6: 必须选择 SW 选项

五线连接10MHz最快，三线时1MHz更稳定

Debug
Connect & Reset Options
Connect: Normal Reset: SYSRESETREQ
☑ Reset after Connect
☐ Log Debug Accesses
Cache Options
☑ Cache Code
☑ Cache Memory
Download Options
☐ Verify Code Download
☐ Download to Flash

7: 必须用 SYSRESETREQ 选项!!! 重要项!!!



Options for Target '???'

○ Use Simulator with restrictions ● Use: CMSIS-DAP Debugger

CMSIS-DAP Cortex-M Target Driver Setup

Erase Sectors 只擦除数据所在扇区

Download Function
○ Erase Full Chip ☑ Program
● Erase Sectors ☑ Verify
○ Do not Erase ☑ Reset and Run

RAM for Algorithm
Start: 0x20000000 Size: 0x00001000

Reset and Run 下载后自动复位

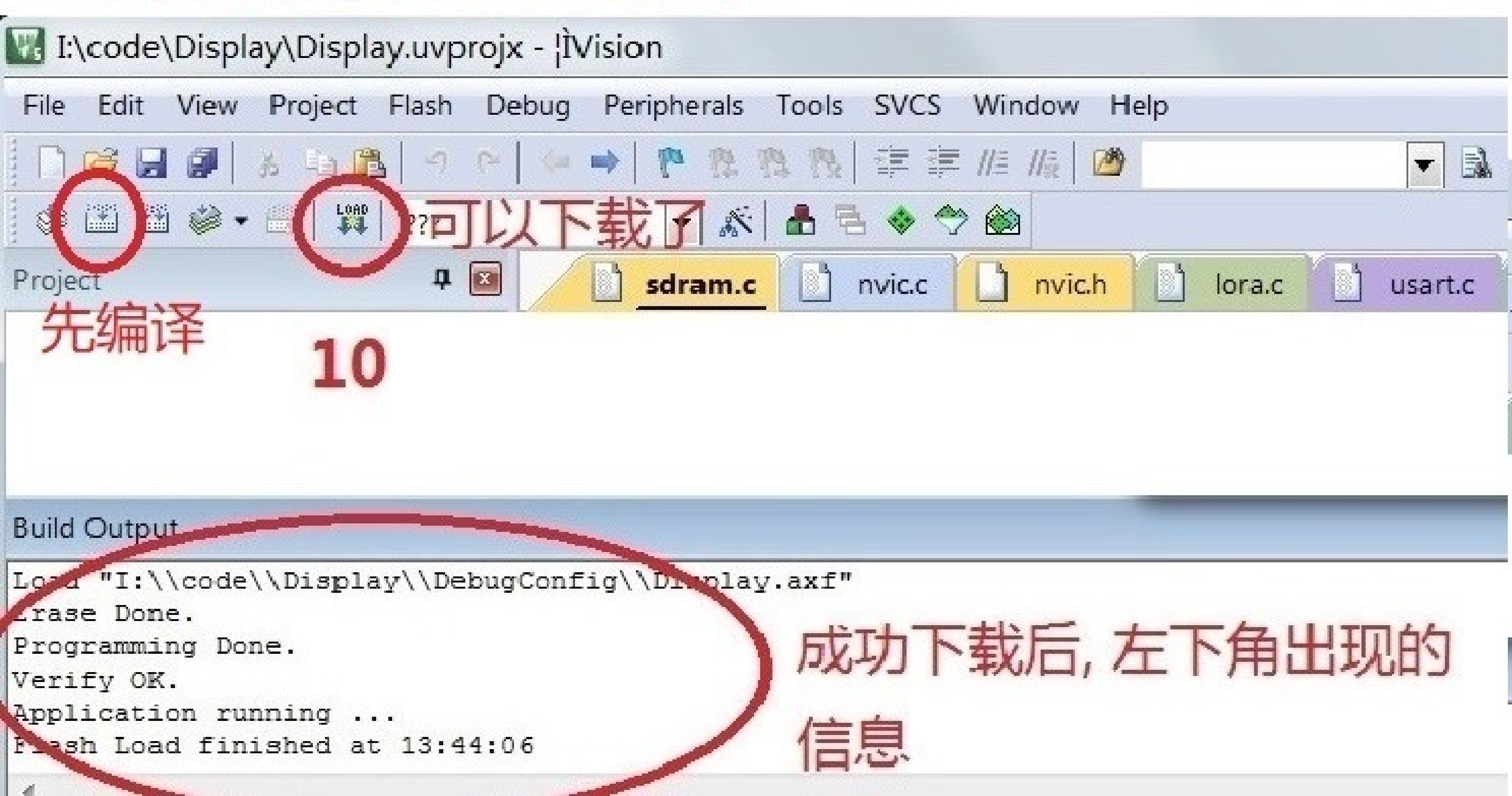
Programming Algorithm

| Description | Device Size | Device Type | Address Range |
|---|---|---|---|
| STM32F4xx Flash | 1M | On-chip Flash | 08000000H - 080FFFFFH |

ADD按钮 可更换目标芯片FLASH大小；
尽量不要在这里更改，只要上面第3步选择正确就可以

Start: Size:

# 3: 代码烧录 (先编译,后下载)



I:\code\Display\Display.uvprojx - µVision

可以下载了

先编译

```
Load "I:\\code\\Display\\DebugConfig\\Display.axf"
Erase Done.
Programming Done.
Verify OK.
Application running ...
Flash Load finished at 13:44:06
```

成功下载后，左下角出现的信息